MILITARY SPECIFICATION

CABLE ASSEMBLY, SPECIAL PURPOSE, ELECTRICAL: W56 and CABLE, TRAINING, SPECIAL PURPOSE, ELECTRICAL: W56

This specification is approved for use by the US Army Armament, Munitions, and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers two types of electrical cable assembies, designated Cable Assembly, Special Purpose, Electrical: W56 and Cable, Training, Special Purpose, Electrical: W56. These cables are used with the Nike Hercules warhead section. The W56 cable assembly is used to simulate warhead section circuits during ground and pre-launch operations and is removed prior to launch.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS), specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAN-Q(D), Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

8808362 - Packing and Marking for Storage and Shipments.
8883539 - Bag, Packing, for Repair Parts.
9268620 - Cable Assembly, Special Purpose, Electrical: W56.
9268621 - Cable, Training, Special Purpose, Electrical: W56.

Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components. Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on component drawings. The contractor shall have available verifiable proof (i.e., objective evidence (see 6.2.1)) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly. The W56 cable assemblies shall comply with all requirements specified on drawings (dwg.) 9268620 and 9268621, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 First article. When specified, sample cable assemblies shall be subjected to first article inspection (see 4.3 and 6.1.1).

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3.4 Continuity and circuit resistance. The cable assembly shall exhibit continuity between connector terminals, as specified in Table I (as applicable), and the resistance shall be not greater than 0.10 ohm. The circuit resistance of the cable assembly shall be as specified in Table II. Testing shall be as specified in 4.5.1.

TABLE I. Continuity.

dwg. 9268620		dwg. 9268621	
From	To	From	To
J1-B	J1-C	J1-D	J1-G
B	L	D	H
B	Z	D	W
D	G	D	X
D	H	D	J8-C
D	T	D	K
D	U	D	M
D	W	D	P
D	X	D	U
D	J8-C	D	V
D	K	D	Y
D	M	E	J1-V
D	U	E	J8-A
D	V	N	F
D	Y	N	J1-P
E	J1-V		
E	J8-A		
F	R		
N	F		
N	J1-P		
Y	J8-P		

TABLE II. Circuit resistance.

dwg. 9268620 and 9268621		
From	To	Resistance (ohms ± 3%)
J8-C	J8-H	511
C	J	511
C	L	511
C	S	511
C	T	511
C	Z	511
C	a	511

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3.5 Insulation resistance. The insulation resistance of the cable assembly shall be not less than 20 megohms. After humidity testing, the insulation resistance of the cable assembly shall be not less than 15 megohms. There shall be no evidence of insulation breakdown or other damage during this test. Testing shall be as specified in 4.5.2.

3.6 Dielectric strength. The cable assembly shall be subjected to a high-potential alternating current at a frequency of 60 hertz. The cable assembly shall show no indication of dielectric failure as evidenced by flashover (surface discharge), sparkover (air discharge), breakdown (puncture discharge), or inability to attain or maintain the high potential. Testing shall be as specified in 4.5.3.

3.7 High temperature. The cable assembly shall withstand a high temperature of 165° ± 5°F, for a minimum period of 4 hours, without evidence of cracking, swelling, distortion, or other damage. After this test (at room temperature), the cable assembly shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.4.

3.8 Low temperature. The cable assembly shall withstand a low temperature of -33° ± 3°F, for a minimum period of 4 hours, without evidence of cracking, swelling, distortion, or other damage. After this test (at room temperature), the cable assembly shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.5.

3.9 Humidity. The cable assembly shall withstand a relative humidity of 95 ± 5 percent and a temperature of 105° ± 5°F without evidence of cracking, swelling, distortion, or other damage. After this test (at room temperature), the cable assembly shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.6.

3.10 Serialization. A serial number shall be assigned by the contractor to each cable assembly produced. In no event, shall serial numbers be duplicated, nor shall they give any indication of classified production quantities. Six-letter serial codes (e.g., VIN SAR) shall not be used on these cables.

3.11 Soldering. All soldering of the cable assemblies shall be in accordance with MIL-STD-1460.

3.12 Workmanship. All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection, the contractor shall submit two consecutively produced W56 cable assemblies, of each type being produced, to the testing facility designated in the invitation for bids or request for proposal.

4.3.2 Inspection and testing to be performed. The first article inspection samples shall be inspected for the defects specified in 4.4.2.1, using inspection methods contained therein, and for all the requirements of Section 3. The test requirements of Section 3 are listed in Table III, and the cable assemblies shall be tested using the test methods specified in 4.5. Inspection and testing shall be performed with the acceptance inspection equipment specified in 4.4.4. The cable assemblies shall be tested in the order specified in Table III. If any cable assembly fails to comply with any of the applicable requirements, the first article samples shall be rejected. The Government reserves the right to terminate its inspections upon any failure of the first article samples to comply with the stated requirements. This series of tests is considered destructive, and samples so tested shall not be returned to the lot.

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TABLE III. First article tests.

Test	Requirement paragraph
a. Continuity and circuit resistance	3.4
b. Insulation resistance	3.5
c. Dielectric strength	3.6
d. High temperature	3.7
e. Low temperature	3.8
f. Humidity	3.9

4.4 Quality conformance inspection.

4.4.1 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.4.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table IV). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

TABLE IV. Inspection lot sampling.

Major defect table		Minor defect table	
Lot size	Sample size	Lot size	Sample size
1-5	all	1-5	all
6-10	5	6-10	4
11-15	10	11-15	8
16-25	15	16-25	11
26-35	22	26-35	14
36-50	26	36-50	16
51-70	30	51-70	18
71-100	33	71-100	19
101 and over	37	101 and over	20

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4.4.2.1 Classification of defects. Critical, major, and minor defects are defined in MIL-STD-105.

4.4.2.1.1 Cable Assembly, Special Purpose, Electrical: W56 (see dwg. 9268620) and Cable, Training, Special Purpose, Electrical: W56 (see dwg. 9268621).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Length, max.	Gage
102.	Ability of connector to engage and disengage with mating receptable (2 connectors)	Manual/Gage
103.	Connector improper or damaged (including missing, loose, or damaged connector terminals (pins or sockets) and damaged insert material) (2 connectors)	Visual/Manual
104.	Keyslot orientation of connector improper (2 connectors)	Visual
105.	Depth of socket or height of pin in connectors (see 6.4)	Visual
106.	Soldering improper (inspect during assembly) (see 3.11)	Visual
Minor:		
201.	Cable jacket cut or damaged (exposing inner conductors)	Visual
202.	Connector cover missing, damaged, or improper (including chain and retaining washer) (2 places)	Visual
203.	Potting missing or improper (2 places)	Visual
204.	Serialization missing, improper, or illegible (see 3.10)	Visual
205.	Marking missing, improper, or illegible (including pin or socket designations on connector insert material)	Visual
206.	Evidence of poor workmanship (see 3.12)	Visual

4.4.2.1.2 Packaging of Cables (see Section 5).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Cable improperly preserved, packaged, packed, and marked (see applicable paragraph in Section 5)	Visual

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Minor:
201. Evidence of poor workmanship (see 3.12) . . . . Visual

4.4.3 Quality conformance inspection testing. The contractor shall perform all of the tests specified in Table V. The tests shall be performed in the listed sequence. None of these tests is considered destructive, and samples so tested shall be returned to the lot.

TABLE V. Quality conformance tests.

Test	Test classification
a. Continuity and circuit resistance	Major
b. Insulation resistance	Major

4.4.3.1 Continuity and circuit resistance (see 3.4) and insulation resistance (see 3.5). Each cable assembly shall be subjected to these tests (see applicable paragraph in 4.5). If any cable assembly fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration procedures designed or specified by the contractor (see 6.1.2). The contractor shall obtain approval of such equipment designs, operating instructions, and calibration procedures prior to use on the contract (see 6.3).

4.4.4.2 Minor defects and tests. Not applicable.

4.5 Methods of inspection. The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, the tests shall be conducted at room temperature (77° ± 10°F). NOTE: During environmental tests, cable connectors shall be protected by mating connectors (e.g., test connectors) or by suitable protective caps.

4.5.1 Continuity and circuit resistance. The continuity and circuit resistance of the cable assembly shall be measured by use of a Government-approved resistance measuring device which is capable of accurately measuring resistances in the ranges specified in Tables I and II. The cable assembly shall comply with the requirements of 3.4 during this test.

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4.5.2 Insulation resistance. The insulation resistance of the cable assembly shall be measured by using a Government-approved insulation resistance tester at an applied voltage of 500 ± 50 volts direct current. The measurements shall be made between each connector terminal (pin or socket) and all other electrically isolated terminals and connector shells tied together as one terminal (refer to the cable drawing for the wiring diagram). The test voltage shall be applied instantaneously, and the insulation resistance shall be measured after a 10-second application of the test voltage. The cable assembly shall comply with the requirements of 3.5 during this test.

4.5.3 Dielectric strength. The cable assembly shall be subjected to a potential of 500 ± 50 volts rms for a minimum period of 10 seconds. The test potential shall be applied instantaneously between each connector terminal and all other electrically isolated terminals and connectors shells tied together as one terminal (refer to the applicable cable drawing for the wiring diagram). A Government-approved test setup shall be used, and suitable means shall be provided to indicated the occurrence of dielectric failure in the event that it is not visually evident in the cables. The cable assembly shall comply with the applicable requirements of 3.6 during this test.

4.5.4 High temperature. The cable assembly shall be placed into a test chamber, and the temperature of the chamber shall be raised to 165°F. The cable shall remain at this temperature for a minimum period of 4 hours. After this high-temperature exposure period, the cable shall be removed from the test chamber, allowed to stabilize to room temperature (without the use of forced cooling), and then tested as specified in 4.5.1, 4.5.2, and 4.5.3. The cable assembly shall comply with the requirements of 3.7 during this test.

4.5.5 Low temperature. The cable assembly shall be placed into a test chamber, and the temperature of the chamber shall be lowered to -33°F. The cable shall remain at this temperature for a minimum period of 4 hours. After this low-temperature exposure period, the cable shall be removed from the test chamber, allowed to stabilize to room temperature (without the use of forced heating), and then tested as specified in 4.5.1, 4.5.2, and 4.5.3. The cable assembly shall comply with the requirements of 3.8 during this test.

4.5.6 Humidity. The cable assembly shall be placed into a dry oven and conditioned at a temperature of 105°F for a minimum period of 24 hours. At the end of this conditioning period, the cable shall be transferred to a humidity chamber having a temperature of 105°F and a relative humidity of 95 percent. The cable shall remain in this chamber for a minimum period of 96 hours.

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After this exposure period, the cable shall be removed from the humidity chamber, allowed to stabilize to room temperature (without the use of forced cooling), and then tested as specified in 4.5.1, 4.5.2, and 4.5.3. The cable assembly shall comply with the requirements of 3.9 during this test.

5. PACKAGING

5.1 Cable assemblies for interplant shipments (see 6.2.2).

5.1.1 Preservation, packaging, packing, and marking. The cable assemblies shall be preserved, packaged, packed, and marked in accordance with MIL-STD-1188 or as specified in the contract or purchase order.

5.2 Cable assemblies for spare (repair) parts.

5.2.1 Preservation, packaging, and marking. Preservation, packaging, and marking shall be in accordance with dwg. 8883539 or as specified in the contract or purchase order.

5.2.2 Packing and marking. Packing and marking shall be in accordance with dwg. 8808362 or as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Whether cable procurement is for spare (repair) parts or for interplant shipments (see 6.2.2).
c. Applicable methods of packaging and packing (see Section 5).
d. First article sample requirements.

6.1.2 Contract data requirements. Contractor engineering documentation shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) when specified on a Contract Data Requirements List (DD Form 1423) incorporated into the contract or purchase order. Deliverable data required by this specification is cited in the following paragraph:

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Paragraph No.	Data requirement	Applicable DID No.
4.4.4.1	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation.	DID-R-1714 (Tailored)

Copies of data item descriptions, required by the contractor in connection with specific acquisition functions, should be obtained from the procuring activity or as directed by the contracting officer.

6.2 Definitions.

6.2.1 Objective evidence. Records of the contractor's quality control and inspections which can be verified.

6.2.2 Interplant shipments. Shipments of an item from a manufacturer, or from a supply source, to a receiving activity for (a) use of the item in the next higher assembly (during end item production) or (b) for repackaging of the item, in accordance with 5.2, for use as spare (repair) parts.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration procedures from Commander, US Army Armament Research and Development Center, Attn: DRSMC-QAN-I(D), Dover, New Jersey 07801.

6.4 Inspection of connectors.

6.4.1 Height of pins. In the event a pin does not appear to be the same relative height of adjacent pins, a gage shall be used to determine the variation from normal height. Reject any connector that has a pin that varies by more than 0.060 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be used as the reference surface for all measurements.

6.4.2 Depth of sockets. In the event a socket does not appear to be the same relative depth of adjacent sockets, a gage shall be used to determine the variation from normal depth. Reject any connector that has a socket that varies by more than 0.060 inch maximum from the normal socket depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward metal edge of the connector shall be used as the reference surface for all measurements.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading of US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

Custodian:
Army - AR

Preparing activity:
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(Project 1135-A253)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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